

GABRIELE GIACOMETTI

MACHINE LEARNING ENGINEER · PHYSICS PHD

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PROFILE

Machine Learning Engineer with a Theoretical Physics PhD background, focused on taking models from research to deployed, monitored services. End-to-end experience building NLP, LLM/RAG, and time-series systems with reproducible pipelines, REST APIs, serverless cloud deployment, and CI/CD (FastAPI, Docker, GCP Cloud Run, GitHub Actions, Terraform). Strong Python/PyTorch engineering and rigorous experimentation. Available to start immediately.

TECHNICAL SKILLS

Languages: Python, SQL, LaTeX, Mathematica

ML / DL: PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers, XGBoost

LLM / NLP: LangGraph agents, MCP, RAG, LangChain, Amazon Bedrock, sentence-transformers, ChromaDB, FAISS, Ollama, RAGAS, DistilBERT fine-tuning

Cloud / MLOps: AWS (Lambda, Bedrock, DynamoDB, ECR), GCP (Cloud Run, Artifact Registry), Terraform (IaC), GitHub Actions CI/CD, Docker, Cloud Logging & Monitoring, Workload Identity Federation

Backend / Data: FastAPI, REST APIs, Streamlit, Git, Pandas, NumPy, SciPy, time-series (LSTM, Prophet, PatchTST)

PROJECTS

Agentic RAG on AWS — evaluated LLM agent, deployed with Terraform — *LangGraph · MCP · Amazon Bedrock · AWS Lambda · DynamoDB · Terraform* github.com/lele25896/agentic-rag-aws

- Built a LangGraph agent with four tools (paper retrieval, structured PDF extraction, sandboxed calculator, web search), human-in-the-loop approval via interrupt() with a DynamoDB checkpoint, and the same tools exposed through an MCP server.
- Compared ReAct and plan-and-execute designs on a 30-task eval (task success, tool choice, latency, cost) with Claude Haiku 4.5 on Bedrock and a local Gemma 12B: equal quality (87% vs 90%), plan-and-execute at 2x the cost and latency.
- Deployed on AWS with Terraform (Lambda container, Function URL, DynamoDB, Bedrock, budget alarm) and fixed the failures only the real cloud exposed (Bedrock schema limits, Function URL permissions, base64 bodies); a full 30-task eval costs under USD 1.

Fraud Detection API — cloud-deployed MLOps pipeline — *PyTorch · FastAPI · Docker · GCP Cloud Run · GitHub Actions* github.com/lele25896/Credit-Card-Fraud-Detection-with-Autoencoders

- Built an unsupervised autoencoder anomaly detector on imbalanced transaction data; a weighted reconstruction score raised F1 on the fraud class from 0.56 to 0.81.
- Containerized the model as a FastAPI REST service and deployed it serverless on GCP Cloud Run, with a public Streamlit dashboard for end-to-end inference.
- Automated the pipeline with GitHub Actions CI/CD (pytest, build, push to Artifact Registry, deploy) via keyless Workload Identity Federation and Terraform IaC (remote GCS state, plan-gated PRs), plus structured Cloud Logging + uptime monitoring.

Local RAG Assistant for Scientific Papers — *LangChain (LCEL) · ChromaDB · Ollama (Llama 3) · RAGAS* github.com/lele25896/langchain-rag-assistant

- Built a fully local, corpus-agnostic Retrieval-Augmented Generation system for Q&A over scientific PDFs, composed end-to-end with LangChain LCEL (RunnableParallel/RunnablePassthrough) — HuggingFace embeddings on GPU, Chroma vector store, ChatOllama generation with inline citations.
- Evaluated with a RAGAS suite (faithfulness 0.75, answer relevancy 0.77, context recall 0.70) on an embedding/chunking config (all-mpnet-base-v2 @ chunk 512) selected via a prior 3x3 ablation.

News Category Classifier — *PyTorch · HuggingFace Transformers · Gradio* github.com/lele25896/news-category-classifier

- Fine-tuned DistilBERT for multi-class text classification over ~210k articles (10 categories), reaching Macro F1 0.842 (+9.5pp over a TF-IDF baseline).
- Deployed as a live interactive demo on HuggingFace Spaces.

Energy Consumption Forecasting — *PyTorch · XGBoost · Prophet · PatchTST · Streamlit* github.com/lele25896/Energy-Consumption-Forecasting

- Benchmarked 4 forecasting models (Prophet, XGBoost, LSTM, PatchTST) on 121k hours of real electricity-demand data under a realistic autoregressive rollout (672h), exposing how model rankings invert versus misleading teacher-forced evaluation.
- Best model XGBoost + lag features: MAE 983 MW, MAPE 6.40% in production-realistic rollout; diagnosed LSTM exposure bias and halved its error (27%→14% MAPE) via scheduled sampling; delivered an interactive Streamlit dashboard for scenario analysis.

EXPERIENCE

Research Intern · National Centre for Nuclear Research (NCBJ), Warsaw 2025

- Modeled complex non-linear systems with analytical and algorithmic methods; authored a peer-reviewed scientific publication and presented results to international audiences.

PhD Candidate in Theoretical Physics · Università di Catania 2023-2026

- Develop mathematical models for complex systems and implement large-scale numerical simulations in Python (NumPy/SciPy); collaborate across international research teams.

Senior Tutor — Mathematical Methods for Physics · Università di Catania 2024

- Designed and delivered structured quantitative problem-solving sessions for STEM students.

EDUCATION

PhD in Theoretical Physics · Università di Catania 2023-2026

MSc in Physics · Università di Catania · 108/110 2019-2023

BSc in Physics · Università di Catania · 96/110 2015-2019

Publications: 3 peer-reviewed papers (theoretical / quantum gravity). **Languages:** Italian (native), English (fluent), Korean & Spanish (basic).